Inflammation Summary Report

Below is an overview of your genes in relation to immune response. It is offered for educational purposes only. Please see the linked articles for details and complete references.

Article: TNF-alpha

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TNF	rs1800629	GG	A	0.15	Increased TNF alpha, increased risk of many chronic inflammatory diseases
TNF	rs361525	GG	A	0.05	Increased TNF alpha, increased risk of many chronic inflammatory diseases
TNF	rs1799964	TC	C	0.21	Increased TNF alpha, increased risk of many chronic inflammatory diseases
TNF	rs1799724	CC	T	0.12	Increased TNF alpha, increased risk of many chronic inflammatory diseases
TNFRSF1A	rs1800693	CT	C	0.39	Increased risk of multiple sclerosis; increased NF-kB signaling
TNFRSF1A	rs767455	CT	C	0.42	Increased risk of inflammatory diseases.
TNFRSF1B	rs1061622	TT	G	0.23	Increased risk of psoriasis, lupus
TNF	rs1800610	GG	A	0.08	Lower TNF; less inflammation but more susceptible to infectious diseases

Article: IL-17

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL17A	rs2275913	AA	A	0.33	Increased risk of autoimmune, periodontal, and bowel disease
IL17A	rs279548	CC	T	0.24	Somewhat increased IL17A, increased risk of asthma, atopy
IL17F	rs763780	TT	C	0.05	Increased risk of rheumatoid arthritis
IL17A	rs8193037	--	A	0.01	Decreased risk of some inflammatory conditions, decreased IL-17A
IL17F	rs3819025	--	A	0.06	Decreased risk of some inflammatory conditions, decreased IL-17A

[Article: Inflammation Mood]

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL6	rs1800797	GG	A	0.36	Increased depression risk (Chinese pop.)
IL6R	rs4129267	CC	C	0.62	CC only: Increased risk of anxiety, depression
IL1B	rs16944	AG	G	0.63	GG only: Increased IL1B; Increased risk depression
IDO1	rs9657182	TT	C	0.41	CC only: more likely to have depr. with inflammation

[Article: NLRP3]

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CIAS1	rs35829419	CC	A	0.03	Increased NLRP3 activation
CIAS1	rs1539019	CC	A	0.38	AA only: Increased NLRP3
CIAS1	rs10754558	--	C	0.61	Somewhat increased NLRP3 activation
CIAS1	rs3806265	--	C	0.34	Somewhat increased NLRP3 activation
CIAS1	rs10733113	--	A	0.15	Somewhat increased NLRP3 activation
CIAS1	rs12048215	--	G	0.11	Somewhat increased NLRP3 activation
CIAS1	i5007539	--	G	0	Carrier of a mutation linked to familial cold urticaria
CIAS1	rs28937896	--	C	0	Carrier of a mutation linked to familial cold urticaria
CIAS1	rs121908147	GG	A	0.007	Carrier of a mutation linked to familial cold urticaria
CIAS1	rs121908150	--	T	0	Carrier of a mutation linked to familial cold urticaria
CIAS1	rs121908148	--	G	0	Carrier of a mutation linked to familial cold urticaria
CIAS1	i6015531	--	A	0.03	Increased NLRP3 activation
CIAS1	rs7525979	CC	T	0.06	increased risk of cognitive impairment in aging; increased risk of gouty arthritis (TT only)
CIAS1	rs4925648	TC	T	0.1	increased risk of asthma; TT- increased risk of Crohn's

Article: Back Pain

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL1A	rs1800587	AG	A	0.28	Increased inflammation, pain from degenerative disc disease
IL6	rs1800795	GG	C	0.34	C/C: less risk of disc degeneration
CILP	rs2073711	AA	A	0.38	Decreased risk of disc disease
COL1A1	rs1800012	CC	A	0.16	Increased risk of pain from degenerative disc disease
COL2A1	rs2276454	--	G	0.6	Increased risk of pain from degenerative disc disease
COL11A1	rs1676486	AG	A	0.18	Increased risk of pain from degenerative disc disease
CASP9	rs4645978	--	C	0.44	Increased risk of pain from degenerative disc disease
PARK2	rs926849	TT	C	0.32	Increased risk of pain from degenerative disc disease

Article: Gingivitis

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TNF	rs1800629	GG	A	0.15	Increased risk of inflammatory gum diseases
IL1A	rs1800587	AG	A	0.28	Increased risk of inflammatory gum diseases
IL1B	rs1143634	GG	A	0.22	Increased risk of inflammatory gum diseases
IL6	rs1800795	GG	C	0.34	Increased risk of inflammatory gum diseases
IL8	rs4073	AT	A	0.5	Increased risk of inflammatory gum diseases
IL10	rs1800896	TT	C	0.45	C/C: higher IL-10, less likely to have gum disease (good)
CCR5	i3003626	--	D	0.09	Decreased risk of periodontal disease
IL2	rs2069763	CA	A	0.31	3x increased risk of chronic periodontitis

[Article: Chronic Inflammation]

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TNF	rs1800629	GG	A	0.15	Higher TNF levels, increased risk of many chronic inflammatory conditions
TNF	rs361525	GG	A	0.05	Higher TNF levels, increased risk of many chronic inflammatory conditions
IL8	rs4073	AT	A	0.5	A/A: Increased IL8; increased risk of periodontitis, gastritis, Alzheimers, diabetic nephropathy
IL6	rs1800795	GG	C	0.34	C/C: lower risk of gingivitis
IL1B	rs16944	AG	G	0.63	G - Typical risk of septic shock; A/A: Increased risk of septic shock
IL1B	rs1143634	GG	A	0.22	Increased risk of gingivitis
IL1A	rs1800587	AG	A	0.28	Increased IL1A, increased risk of gum disease, tinnitus, acne, hearing loss
IL10	rs1800896	TT	C	0.45	CC: higher IL-10 (usually good!)
NLRP3	rs35829419	CC	A	0.03	Increased susceptibility to several chronic inflammatory diseases.
HMGB1	rs1045411	CC	T	0.23	increased sepsis risk, higher HMGB1 levels in infection;
INFG	rs2430561	--	A	0.39	increased inflammatory and sickness behavior
MTHFR	rs1801133	GG	A	0.33	Decreased methyl group production, decreased detoxification of mercury and arsenic, possibly decreased melatonin production
GSTM1	rs366631	--	A	0.78	AA: GSTM1 null, increased risk of cancer, increased negative effects of smoking
GSTO1	rs4925	CC	A	0.3	Decreased detoxification of arsenic; increased risk of PCOS
GSTA1	rs3957357	--	A	0.38	Decreased detoxification, increased risk of depression,
NFE2L2	rs6721961	GG	T	0.11	Decreased Nrf2, increased risk of male infertility, increased risk of CVD,
AS3MT	rs11191439	TT	C	0.1	Arsenic is more harmful
NQO1	rs1800566	AG	A	0.2	Increased risk of cancer from benzene and smoking, increased risk of Parkinson’s from pesticide exposure
SOD1	rs1041740	CC	T	0.3	Increased ROS, increased risk of kidney problems, heart disease
SOD2	rs5746136	CT	T	0.28	Increased ROS, increased risk of asthma, PCOS

[Article: Mast Cells]

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
KIT	i5007903	--	T	0.00005	KIT D816V mutation (see article for caveats)
KIT	rs121913507	--	T	0.00005	KIT D816V mutation (see article for caveats)
IL-13	rs1800925	TC	T	0.16	Increased risk of systemic mastocytosis, rhinitis, asthma
IL4R	rs1801275	AA	A	0.74	better prognosis in systemic mastocytosis
FCER1A	rs2298805	GG	A	0.003	Decreased risk of hives, lower IgE response
FCER1A	rs2251746	TT	C	0.25	Decreased IgE response
FCER1A	rs2427827	TC	T	0.4	Increased IgE, increase sinus problems, allergic reactions
CMA1	rs1800875	--	T	0.46	Decreased IgE, lower risk of a-fib
PTPN22	rs2476601	GG	A	0.08	Increased psoriasis, arthritis, T1D, lupus, urticaria risk
IL33	rs1342326	AA	C	0.17	Increased risk asthma, hay fever
IL33	rs3939286	CT	T	0.26	Increased risk of asthma
IL33	rs928413	AG	G	0.28	Increased risk hay fever, asthma
ALDH2	rs671	GG	A	0.006	Increases mast cell activation
PTGS2	rs4140564	--	G	0.05	Increased risk osteoarthritis
AOC1	rs10156191	CT	T	0.26	Reduced production of DAO, increased risk of migraines due to histamine
AOC1	rs2052129	GT	T	0.23	Reduced production of DAO, increased risk of migraines due to histamine
AOC1	rs1049742	--	T	0.07	Reduced production of DAO
AOC1	rs1049793	CC	G	0.31	Reduced production of DAO
HNMT	rs1050891	GA	A	0.79	Reduced breakdown of histamine compared to G/G
HNMT	rs11558538	CC	T	0.1	Reduced HNMT activity, reduced breakdown of histamine
HNMT	i3000469	--	T	0.1	Reduced HNMT activity, reduced breakdown of histamine

Article: ME/CFS

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PTPN22	rs2476601	GG	A	0.08	increased risk of autoimmune diseases, increased susceptibility to CFS/ME (in patients with infectious disease onset)
CTLA4	rs3087243	GG	G	0.57	increased risk of autoimmune conditions, decrease CTLA4 expression; increased risk of CFS/ME (patients with infectious disease onset only)
TNF	rs1799724	CC	T	0.12	higher TNF-alpha levels, increased susceptibility to CFS/ME
INFG	rs2430561	--	A	0.39	decreased risk of ME/CFS
NLRP3	rs35829419	CC	A	0.03	Increased susceptibility to fatigue / pain after EBV or other viruses
NLRP3	rs121908147	GG	A	0.007	mutation linked to autoinflammatory disease in combo with other genes (rare)
TRPM8	rs11563204	AG	A	0.2	increased risk of CFS/ME (cold, menthol receptor)
TRPM3	rs6560200	--	C	0.5	CC: common genotype, higher risk of CFS/ME (Naltrexone may work?)
TRPM3	rs1891301	--	T	0.51	TT: higher risk of CFS/ME
CFB	rs4151667	TT	A	0.03	AA: increased risk of CFS/ME
CFH	rs1061170	--	C	0.37	CC: increased risk of AMD, higher immune response; TT: increased risk of CFS/ME

Article: Hearing Loss

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
EDN1	rs5370	TT	T	0.21	Increased risk of sudden sensorineural hearing loss (related to vasoconstriction)
F5	rs6025	CC	T	0.02	Factor V Leiden (clot related), increased risk of SSNHL
MTHFR	rs1801133	GG	A	0.33	Increased risk of SSNHL (folate-related)
SOD1	rs4998557	--	A	0.19	A/A only: increased relative risk of SSNHL
IL1R2	rs4141134	--	G	0.41	Increased risk of SSNHL
UCP2	rs659366	CT	T	0.37	Increased risk of SSNHL
IL6	rs1800796	CC	G	0.92	Higher IL-6; increased risk of SSHL
HSP70	rs2763979	CT	T	0.35	Increased risk of noise-induced hearing loss
CYP1A1	rs1799814	GG	T	0.04	Increased CYP1A1 activity, increased risk of SSNHL

Article: Acne

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TYK2	rs33980500	CC	T	0.07	Decreased severe acne risk
TNF	rs1800629	GG	A	0.15	3-fold Increased acne risk
TNF	rs1799724	CC	T	0.12	Decreased acne risk
CTLA4	rs3087243	GG	G	0.57	G/G: higher risk of acne
IL1A	rs1800587	AG	A	0.28	Increased acne risk (inflammation)
IL1A	rs17561	AC	A	0.28	Increased acne risk (inflammation)
IL6	rs1800796	CC	G	0.07	Increased acne risk (inflammation)
RETN	rs3745367	GG	A	0.26	Increased acne (sebum)
BCO1	rs7501331	--	T	0.21	Decreased beta-carotene conversion.
BCO1	rs12934922	AA	T	0.41	decreased beta-carotene conversion
CYP17A1	rs743572	GG	G	0.39	Increased risk of acne (hormones)
HSD11B1	rs846910	GG	A	0.04	A/G: 4-fold increase in acne risk
LCT	rs4988235	GG	G	0.49	GG only: Increased acne risk with dairy
TGFB1	rs1159268	AA	A	0.32	slight increase in the risk of acne
TGFB1	rs38055	GG	A	0.31	slight increase in the risk of acne
WNT10A	rs121908120	TT	A	0.02	protective against acne

Article: Specialized Pro-Resolving Mediators

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FADS1	rs174546	CC	T	0.32	Lower FADS1 enzyme activity, benefit more from direct EPA/DHA intake
FADS2	rs1535	AA	G	0.32	Lower FADS1 enzyme activity, benefit more from direct EPA/DHA intake
ALOX5	rs4987105	CT	T	0.13	Decreased risk of type 2 diabetes, lower levels of C-reactive protein (good)
ELOVL2	rs3734398	TC	C	0.44	Decreased conversion of EPA to DHA
ALOX5AP	rs17216473	--	A	0.06	Increased risk of heart attack (population eating Western / high omega-6 diet)
ALOX12	rs1126667	GA	A	0.41	Slightly decreased risk of breast cancer, lower blood pressure (study group eating Western/ high omega-6 diet)
COX2	rs4648310	--	C	0.01	Low DHA/EPA intake associated with a significantly increased risk of prostate cancer, but high DHA/EPA ameliorates the increased risk
COX2	rs5275	AA	G	0.34	Increasing intake of EPA/DHA reduces prostate cancer risk by 70%
GPR18	rs3742130	GG	A	0.21	SPM receptor; alters risk of IBD
CMKLR1	rs1878022	--	C	0.34	Increased resolvin E1 receptor expression, reduced inflammation in obesity
GPR37	rs149031046	GG	A	0.008	Protectin D1 receptor mutation; possibly important in autism (rare)
LGR6	rs10920362	--	T	0.44	increased risk of osteoporosis

Article: CBD

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TRPV1	rs8065080	--	C	0.37	less TRPV1 activation
TRPV1	rs161364	--	T	0.28	less TRPV1 activation
TRPV1	rs224534	GG	A	0.35	A/A: less TRPV1 activation
HTR1A	rs6295	GG	G	0.47	G/G: increased HTR1A receptor activity
ADORA2A	rs5751876	TT	T	0.42	T/T: increased anxiety
GPR55	rs3749073	--	A	0.1	A/A: reduced receptor function, increased risk of anorexia

Article: Fatigue

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TNF	rs1800629	GG	A	0.15	Higher TNF-alpha; Curcumin, apigenin, and naringenin may help.
TNF	rs3093662	AA	G	0.07	Higher TNF-alpha; Curcumin, apigenin, and naringenin may help.
IL1B	rs4848306	GA	G	0.56	GG: more fatigue in chronic disease; Dihydromyricetin, quercetin
IL1B	rs1143643	--	T	0.29	More fatigue in chronic illness; Dihydromyricetin, quercetin
IL6	rs1800795	GG	C	0.34	C/C: less fatigue (less IL-6 produced)
INFG	rs2430561	--	A	0.39	Increased fatigue risk when interferon-gamma is elevated
NLRP3	rs35829419	CC	A	0.03	Significantly increased risk of severe fatigue with inflammation; methylene blue, CBD, EGCG

Article: IL13

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
Il13	rs20541	AA	A	0.2	higher IgE levels; higher risk of allergies, allergic rhinitis; increased risk of atopic dermatitis with childhood antibiotic use, increased risk of alopecia areata, increased risk of COPD
Il13	rs1800925	TC	T	0.16	increase IgE levels; increased risk of asthma; increased risk of periodontitis, increased risk of COPD
Il13	rs1295686	TT	T	0.23	increased risk of asthma
Il13	rs848	AA	A	0.22	increased risk of asthma, increased risk of alopecia areata

Article: Tinnitus

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
NFKB1	rs3774937	CC	C	0.28	Faster progression to hearing loss in Meniere's
NFKB1	rs4648011	--	G	0.39	Faster progression to hearing loss in Meniere's
Inter-gen	rs4947296	TT	C	0.07	Increased risk of Meniere's
KCNE1	rs1805127	TC	T	0.36	Increased risk of Meniere's
KCNE3	rs2270676	AA	G	0.1	Increased risk of Meniere's
IL1A	rs1800587	AG	A	0.28	A: lower risk of hearing loss in Meniere's; G/G: (common) Higher risk of sudden hearing loss in Meniere's
ADD1	rs4961	TG	T	0.19	Increased risk of tinnitus
IL1A	rs1800587	AG	A	0.28	Lower risk of tinnitus
SLC12A2	rs10089	--	T	0.2	increased relative risk of tinnitus, especially with mild hearing loss
GRM7	rs11928865	--	A	0.11	increased relative risk of tinnitus

Article: Antibiotic Reactions

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MS4A2	rs569108	GA	G	0.03	G/G: increased risk of eczema with childhood antibiotic use
IL13	rs20541	AA	A	0.2	increased risk of atopic dermatitis with childhood antibiotic use
HLAB	rs114892859	GG	T	0.005	increased risk of penicillin allergy and clindamycin adverse reactions
PTPN22	rs2476601	GG	A	0.08	slightly increased relative risk of liver injury with augmentin
HLA-DR	rs2395029	TT	G	0.03	45 to 80-fold increased risk of drug-induced liver injury with flucloxacillin
HLA-DRB1	rs3135388	GG	A	0.11	increased relative risk of liver injury with augmentin
HLAB	rs9263726	GG	A	0.12	increased relative risk of reaction to sulfamethoxazole (Bactrim)
HLAA	rs2523822	AA	G	0.28	increased relative risk of drug-induced liver disease with augmentin
LGALS3	rs11125	AA	T	0.06	increased risk of beta-lactam antibiotic allergy
GSTM1	rs366631	--	A	0.78	A/A: deletion (null) increased risk of adverse cutaneous reactions to sulphonamides in AIDS

Article: Estrogen and Histamine

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
AOC1	rs10156191	CT	T	0.26	reduced production of DAO
AOC1	rs2052129	GT	T	0.23	reduced production of DAO
AOC1	rs2071514	GG	A	0.19	possibly higher DAO
HNMT	rs1050891	GA	A	0.79	reduced breakdown of histamine compared to G/G
HNMT	rs11558538	CC	T	0.1	reduced HNMT activity, higher histamine levels
HNMT	rs2071048	CC	T	0.58	T/T: increased risk of asthma (and higher histamine), common variant
MTHFR	rs1801133	GG	A	0.33	decreased MTHFR function (C677T allele)
ESR1	rs9340799	GG	G	0.33	G/G: increased risk of endometriosis, likely higher estrogen receptors
GPER1	rs11544331	CC	T	0.23	decreased estrogen receptor activation, lower risk of fibroids
AOC1	rs1049742	--	T	0.07	reduced production of DAO
AOC1	rs1049793	CC	G	0.31	reduced production of DAO

Article: Cold Sores

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
POU5F1	rs885950	AC	C	0.42	increased risk of cold sores
C21orf91	rs10446073	--	G	0.94	increased likelihood of cold sores
C21orf91	rs1062202	--	G	0.31	increased likelihood of cold sores
VDR	rs2228570	--	A	0.38	increased likelihood of cold sores
IL1A	rs1304037	--	T	0.68	increased chance of recurrent cold sores
HCP5	rs4360170	--	G	0.07	increased risk of cold sores

Article: Lymphedema

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ALOX5	rs4987105	CT	T	0.13	Lower levels of inflammation and possibly lower lymphedema, decreased risk of type 2 diabetes
LTA4H	rs1978331	GG	G	0.41	Lower levels of LTA4H, which catalyzes the final step in the synthesis of leukotriene B4
MMP2	rs1030868	--	A	0.37	Higher risk of secondary lymphoma
MMP2	rs2241145	--	C	0.45	Higher risk of secondary lymphoma
FOXC2	rs34221221	--	C	0.44	Increased gene expression (likely higher risk of secondary lymphedema)
TNF	rs1800629	GG	A	0.15	Higher TNF-alpha levels; increased risk of complications with lymphedema
TLR4	rs4986791	CC	T	0.05	Increased risk of complications with lymphedema
VEGFR3	rs10464063	--	T	0.46	Increased risk of secondary lymphedema (study of cancer patients)

Article: COMT in Pain Disorders

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
COMT	rs4680	AG	A	0.48	A/A: Met/Met – slow (40% lower COMT activity); lower pain threshold, higher dopamine; more pain in chronic pain disorders
COMT	rs6267	GG	T	0.003	risk of higher pain sensitivity

Article: Osteoarthritis

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL1RA	rs3917225	--	G	0.41	less likely to have osteoarthritis
IL1B	rs1143634	GG	A	0.22	increased risk of osteoarthritis
TNF	rs1800629	GG	A	0.15	increased TNF-alpha, increased risk of osteoarthritis
IL-17F	rs763780	TT	C	0.05	increased risk of osteoarthritis and RA
HFE	rs1800562	--	A	0.05	increased iron, increased risk of osteoarthritis
VDR	rs1544410	CC	T	0.38	decreased risk of osteoarthritis
VDR	rs731236	AA	A	0.62	increased risk of osteoarthritis
CKM	rs4884	--	G	0.69	decreased risk of osteoarthritis
HIF1A	rs11549465	CC	T	0.1	decreased risk of osteoarthritis
ADAMTS14	rs4747096	--	G	0.16	increased risk of osteoarthritis
ADAM12	rs1871054	--	C	0.51	C/C: increased risk of osteoarthritis
MMP1	rs1799750	--	I	0.48	increased risk of osteoarthritis

Article: Melatonin Immune and Inflammation

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
AANAT	rs28936679	--	A	0.0005	rare, linked to delayed sleep phase disorder
AANAT	rs3760138	--	G	0.52	G/G: decreased melatonin; increased risk of breast cancer, increased risk of lupus
TPH2	rs4290270	AT	T	0.61	increased risk of waking early, increased risk of depression
TPH2	rs4570625	GG	T	0.21	GG: less TPH2 function; G/T and T/T: higher TPH2 Function
MTNR1B	rs10830963	CC	G	0.27	linked to a higher risk of diabetes, increased fasting glucose
MTNR1B	rs1387153	CC	T	0.29	linked to a higher risk of gestational diabetes, increased fasting glucose
MTNR1A	rs2375801	--	C	0.34	increased risk of cancer metastasis
MTNR1A	rs6553010	--	A	0.2	increased risk of cancer metastasis
MTNR1A	rs12506228	CC	A	0.28	likely fewer melatonin receptors in the brain, a greater impact from working the night shift, increased risk of Alzheimer’s

Article: Rosacea

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TNF	rs1800629	GG	A	0.15	higher TNF alpha, increased risk of rosacea
IRF4	rs12203592	CC	T	0.13	increased risk of rosacea
IL13	rs847	--	T	0.21	increased risk of rosacea
HLA-DRA	rs763035	--	A	0.14	increased risk of rosacea
NOD2	rs2066844	CC	T	0.04	increased intestinal permeability, IBD; increased risk of rosacea
VDR	rs731236	AA	A	0.62	lower vitamin D levels; increased relative risk of rosacea with low serum vitamin D levels
VEGF	rs2010963	GG	G	0.63	increased risk of rosacea
MCR1	rs1805007	--	T	0.06	more photoaging, facial aging, increased relative risk of rosacea
HERC2	rs1129038	CC	T	0.62	increased relative risk of rosacea

Article: Mitochondria and light

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYCS	rs1861525	--	G	0.04	decreased rate of cognitive decline in Alzheimer's[ref] (good), interacts with lupus risk
COX1	rs28461189	--	A	0.002	mutation in mitochondrial gene linked to cytochrome c oxidase I deficiency
COX1	rs28679680	--	A	0	mutation in mitochondrial gene linked to cytochrome c oxidase I deficiency **Converted files using WGSExtract may give a false positive here.
TNF	rs1800629	GG	A	0.15	Higher TNF-alpha levels
TNFSF11	rs2277438	AA	G	0.22	G/G: lower BMD, increased RANKL; A/G is typical
TNFSF11	rs2277439	AA	G	0.17	associated with lower BMD, increased RANKL

[Article: Asthma]

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TSLP	rs3806932	GG	G	0.45	increased relative risk of childhood allergic asthma and adult-onset asthma
TSLP	rs1837253	CC	T	0.26	Decreased risk of adult asthma (good)
IL4	rs2243250	CC	T	0.18	increased relative risk of asthma
IL33	rs928413	AG	G	0.28	increased risk of asthma, allergies, and COPD
IL33	rs7037276	--	C	0.06	increased relative risk of asthma
IL33	rs1342326	AA	C	0.16	increased relative risk of asthma
IL33	rs3939286	CT	T	0.26	increased relative risk of asthma
TBX21	rs4794067	CC	C	0.27	increased risk of aspirin-induced asthma, increased risk of asthma in children
TBX21	rs11650354	--	C	0.83	increased risk of asthma
IL17A	rs2275913	AA	A	0.33	Higher IL17A, Increased risk of inflammatory diseases including periodontitis, inflammatory bowel disease, asthma; 3-fold increased risk of COPD (smokers)
IL17A	rs279548	CC	T	0.24	somewhat increased IL17A, increased risk of asthma
TNF	rs1800629	GG	A	0.15	Higher TNF-alpha levels. Increased risk of asthma and COPD
TNF	rs361525	GG	A	0.05	higher TNF-alpha levels, increased risk of asthma, COPD
IFNG	rs2069705	AA	A	0.63	A/A: increased relative risk of asthma; A/G: typical
ILIR1	rs3771166	GG	A	0.37	Decreased risk of asthma
NLRP3	rs10754558	--	C	0.61	higher inflammation, increased relative risk of asthma
IL13	rs20541	AA	A	0.2	higher IgE levels; higher risk of allergies, increased risk of COPD, increased dust mite allergies increased risk of asthma
IL13	rs1295686	TT	T	0.23	increased risk of asthma
IL13	rs1800925	TC	T	0.16	increased IgE levels; increased risk of asthma; increased risk of periodontitis, increased risk of COPD
IL13	rs848	AA	A	0.22	increased risk of asthma, increased risk of alopecia areata
HNMT	rs2071048	CC	T	0.58	increased risk of asthma (and higher histamine), common variant
HRH1	rs901865	TC	T	0.17	increased asthma risk (likely increase HRH1)
MTHFR	rs1801133	GG	A	0.33	two copies of MTHFR C677T, enzyme function decreased by 70 – 80%, increased relative risk of asthma
GSTA1	rs3957357	--	A	0.38	low/ non-functioning enzyme; increased risk of asthma, allergies
GSTP1	rs1138272	CC	T	0.07	Increased risk of asthma and exacerbations with higher air pollution exposure

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GSDMB	rs7216389	TT	T	0.53	Increased risk of childhood asthma 3-fold increased risk of asthma with known mold exposure
GSDMB	rs2305480	GG	G	0.57	increased risk of asthma, especially with cigarette smoke exposure
SOD2	rs5746136	CT	T	0.28	higher phthalate metabolite levels, almost 3-fold increased risk of asthma
CDHR3	rs6967330	GG	A	0.17	higher levels of CDHR3, increased risk of rhinovirus infections and severe asthma in children
ADRB2	rs1042713	GA	A	0.38	more likely to have poor response to long-acting β2-agonist (LABA), increased exacerbations in children treated with LABAs plus inhaled corticosteroids (but not corticosteroids alone)
ARG1	rs2781659	GA	G	0.38	lower acute response to inhaled beta-agonists in children with asthma
HNMT	rs11558538	CC	T	0.1	reduced HNMT activity, higher histamine levels, increased relative risk of asthma
HNMT	i3000469	--	T	0.1	reduced HNMT activity, higher histamine levels, increased relative risk of asthma
HLA-DQ	rs9273349	--	C	0.58	increased relative risk of asthma

Article: DAO Enzyme

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
AOC1	rs10156191	CT	T	0.26	reduced production of DAO, increased risk of migraines due to histamine
AOC1	rs2052129	GT	T	0.23	reduced production of DAO, increased risk of migraines due to histamine
AOC1	rs1049742	--	T	0.07	reduced production of DAO
AOC1	rs1049793	CC	G	0.31	reduced production of DAO (50% reduction)

Article: Fibromyalgia

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
COMT	rs4680	AG	A	0.48	more pain in chronic pain situations, more common in fibromyalgia patients
COMT	rs2097903	--	T	0.51	more common in fibromyalgia patients
HTR3A	rs1062613	CT	T	0.23	increased relative risk of fibromyalgia
ADRB2	rs1042713	GA	A	0.38	increased relative risk of fibromyalgia, increased receptor response to catecholamines
TRPV3	rs395357	CC	T	0.48	higher symptom severity in fibromyalgia
TNF	rs1800629	GG	A	0.15	increased fatigue, pain, and sleep problems in fibromyalgia
TNF	rs1799964	TC	C	0.21	increased pain, anxiety, and depression in fibromyalgia
PLPP6	rs2295963	AA	G	0.18	2-fold increased relative risk of fibromyalgia

Article: Hidradenitis Suppurativa

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TNF	rs361525	GG	A	0.05	higher TNF-alpha levels, increased risk of hidradenitis suppurativa, increased severity in HS
MYDD8	rs6853	AA	G	0.12	G/G: Increased risk of more severe HS
SOX9	rs10512572	GG	A	0.12	increased relative risk of HS
GJB2	rs28931594	--	T	0	carrier of a rare mutation associated with hearing loss and keratitis; increased risk of HS
GJB2	i6011365	--	T	0	carrier of a rare mutation associated with hearing loss and keratitis; increased risk of HS
GJB2	rs104894408	--	A	0.00001	carrier of a rare mutation associated with hearing loss and keratitis; increased risk of HS
GJB2	i5001989	--	A	0.00001	carrier of a rare mutation associated with hearing loss and keratitis; increased risk of HS
PSTPIP1	rs121908130	--	A	0	rare mutation linked to pyoderma gangrenosum, acne, and HS
NCSTN	rs387906896	--	T	0	rare mutation associated with Familial acne inversa 1; found in some HS patients

Article: Resveratrol

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SOD2	rs4880	AA	A	0.51	A/A only: lower SOD2 enzymatic activity, not as much of an anti-inflammatory effect from resveratrol; A/G: intermediate SOD activity, resveratrol lowers inflammatory markers
MTHFR	rs1801133	GG	A	0.33	MTHFR C677T; resveratrol reversed some of the endothelial dysfunction
BDNF	rs6265	CC	T	0.18	TT only: decreased BDNF; in a mouse model of Met/Met, SIRT1 activation with resveratrol reduced inflammation and prevented cardiovascular problems

Article: miRNA

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MIR146A	rs2910164	GC	G	0.75	reduced MiR-146a expression; increased risk of active disease in IBD(autoimmune)
MIR423	rs6505162	AC	A	0.55	~40% reduction in relative risk of esophageal cancer in Caucasians; decreased risk of lung cancer
MIR22	rs11078597	CT	C	0.18	slightly higher triglyceride levels, slightly higher sex hormone binding globulin, slightly lower CRP

Article: Thymosin Beta 4

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TMSB4X	rs850636	GG	A	0.42	lower levels of TMSB4X
ACE	rs4343	AG	G	0.49	G/G: ACE deletion/deletion -- higher levels of ACE enzyme which breaks down the TB4 fragment

Article: Trigger Points

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
COL1A1	rs1800012	CC	A	0.16	increased gene expression of COL1A1; more likely to have pain with tennis elbow; increased risk of back pain (disc degeneration); greater fracture risk in osteoporosis
COL1A1	rs9898186	CC	T	0.16	more likely to have pain with tennis elbow
PDGFRA	rs2228230	--	T	0.15	decreased PDFGA expression
PDGFRA	rs7677751	CC	T	0.13	increased relative risk of astigmatism
COMT	rs9332377	CT	T	0.15	Increased myofascial pain

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